ELEMENT DETECTIVE

Name: ..... **Date:** .....

Part 1: name the element in each riddle. Part 2: write your own riddle. Part 3: circle the best answer.

PART 1: WHO AM I?

1. I have 1 proton and I am the lightest element of all. _____
2. I am the noble gas with atomic number 10, and I glow orange-red in signs. _____
3. I am in group 1 and period 3, I fizz violently in water and my symbol comes from Latin. _____
4. My symbol is Fe and I rust. _____
5. I have 6 protons; diamond, graphite and all living things are made of me. _____
6. I am a halogen with 17 protons, used to keep swimming pools clean. _____
7. I make up about 78% of the air and my atomic number is 7. _____
8. My symbol is Au; I am a shiny yellow metal that never rusts. _____
9. I am the only metal that is a liquid at room temperature. _____
10. I have 2 protons, I am the second-lightest element, and I make party balloons float. _____

PART 2: WRITE YOUR OWN RIDDLE

Magnesium (12):

_____

_____

Oxygen (8):

_____

_____

PART 3: QUICK QUESTIONS

1. Elements in the same period have
   a) the same number of electron shells b) the same number of outer electrons c) the same mass
2. Which of these is a group 2 element?
   a) sodium b) calcium c) aluminium
3. Which pair are in the same group?
   a) lithium and sodium b) carbon and nitrogen c) neon and sodium
4. Which element has the most protons?
   a) potassium b) chlorine c) calcium

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ELEMENT DETECTIVE ANSWER KEY

Name: ..... **Date:** .....

PART 1: WHO AM I?

1. hydrogen 2. neon 3. sodium 4. iron 5. carbon 6. chlorine 7. nitrogen 8. gold 9. mercury 10. helium

PART 2: WRITE YOUR OWN RIDDLE

1. Sample: I have 12 protons, I am in group 2, and I burn with a bright white flame. 2. Sample: I have 8 protons, I am in group 16, and you breathe me in every few seconds.

PART 3: QUICK QUESTIONS

1. a) the same number of electron shells 2. b) calcium 3. a) lithium and sodium 4. c) calcium